

INTERNATIONAL STEM SKILLS ROUND TABLE PHASE III

AI and the STEM Educator

A short report from six meetings on teaching STEM in the age of generative AI

The International STEM Skills Round Table met six times between December 2025 and July 2026 to ask what a teacher and a lecturer have to be in the age of generative AI. Its two earlier phases had described the STEM graduate. This one turned to the people who teach.

International STEM Skills Round Table Phase III. Draft for the steering committee September 2026.

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The full record of the meetings, the source materials and the companion pages are at
<https://avisalmon.github.io/RoundTableIII/>

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1. Introduction

1.1 From the graduate to the educator

Two earlier phases of this Round Table described what a STEM graduate should be able to do. Phase I, over six meetings between May 2022 and February 2023, set out to build “a common language, agreed upon and coherent” for the competencies that excellence in STEM requires, together with the methodologies that develop them and the tools that assess them. It reported in January 2025. Phase II, published two months later, turned those definitions into a practical guide for teachers and lecturers, and added a systematic literature review and a Delphi consensus on how AI in education might support each competency.

Phase I was also candid about why the work was needed. Implementation had been limited and slow, it found, because the definitions in circulation were numerous and often impractical, because accessible teaching and learning methodologies were missing, and because there were no agreed methods or tools for measuring whether a competency had been acquired.

Phase II then named the gap that Phase III inherits. Its own rationale states that educators “must be equipped to serve as mentors in fostering 21st-century STEM skills”. Phase II supplied the materials. It did not describe the person who would have to use them.

Dr. Eli Eisenberg put that to the first meeting of Phase III in a single sentence: the focus has shifted from the profile of the graduates to the profile of the teachers and lecturers.

1.2 Why the question became urgent

Three pressures appear in the record, and only one of them is new.

Generative AI changed what a teacher is for. A student can now ask a machine to explain a concept, solve a problem, write code or draft a laboratory report, and get an answer that is fluent whether or not it is right. The steering team’s assessment was that this is a change of role and not of equipment: the lecturer is no longer the sole source of knowledge but the person guiding a learning process shared with the student and with the machine, and generative AI therefore has to be designed into learning rather than added beside it.

Educators differ from one another far more than policy assumes. Teacher colleges give a full pedagogical training; universities usually give none, so a department contains people trained to teach alongside subject experts with no background in developing competencies at all. The steering record concluded that this forces a reconsideration of recruitment and faculty development policy, and that a differential model matched to an individual’s strengths and field is needed in place of uniform training.

The teacher shortage was reframed as an opening. Raised at the steering session of 16 December 2025 as a cross-cutting problem, it was reconsidered there as an opportunity: to move from a stance of no alternative to a strategic definition of a future educator, to treat recruitment from industry as systemic policy rather than a stopgap, and to build teams in which educators with different strengths complement one another.

Underneath all three sits a finding the steering team recorded before the series began: lecturers want to embed competencies in their teaching and lack the tools to do it, and what they need is accompaniment and professional support.

1.3 The objective, and the standing of this report

The objective was fixed at the steering team's opening session on 19 October 2025 and then repeated, word for word, at the head of every steering summary of the phase:

Formulating the profile of the lecturer and teacher in the era of generative artificial intelligence (GAI), for imparting skills in the STEM subjects in institutions of higher education and in high school.

The steering team also decided what kind of document should come out of it. On 3 February 2026 it agreed that the final product should not be a binding guidelines report or an action recipe, but an integrative document that consolidates the insights into one systemic picture, surfaces the dilemmas, and is meant to inform professional discourse and policy. That is the standing of what follows.

One further requirement was set at the opening session and repeated at every steering meeting afterwards: the recommendations should be practical, and separated by level of impact – personal, institutional and policy. Section 9.3 keeps that separation.

2. Methodology

2.1 How the sequence of dilemmas was set

The six questions were not chosen inside the meetings. They were assembled by the steering team over the first months of the phase and then presented to the room.

The opening session on 19 October 2025 identified three of them: whether the educator must personally possess the capabilities being imparted; whether generative AI should be built into a triangle with the lecturer and the student; and which physical and virtual environments promote the acquisition of competencies. The session of 16 December 2025 added a fourth, and recorded a decision about it that is worth stating plainly – the group declined to settle the question of a uniform or a diverse profile and chose instead to formulate it as the central dilemma for the wider discussion. Professional development was added later still. Prof. Arnon Bentur presented the assembled sequence at the second meeting, and it held from there.

A background document was circulated to participants before each session, prepared by Avi Salmon, Prof. Bentur and Dr. Eisenberg. Two of them survive, for the third and fourth meetings.

2.2 The meetings

Table 1. The six meetings of Phase III, the dilemma put to each, and the invited speakers.

#	Date	Dilemma	Speakers
1	02.12.2025	Convening session. Objective presented; no dilemma put to the room	Avi Salmon (Intel Israel)

2. METHODOLOGY

2	20.01.2026	Should the profiles of lecturers and teachers become uniform, or should there be a diversity of complementary profiles?	Jan Morrison (TIES); Dr. Gabi Shafat (Afeka)
3	10.03.2026	Should educators themselves possess the required competencies, or should their expertise lie in cultivating them in students?	Prof. Sigal Tifferet (Ruppin); Dr. Olena Bekh (ETF)
4	28.04.2026	Should the educator lead the process alone, or should a triangle be formed among educator, student and AI?	Dr. Iris Pinto (Institute for Futures Research in Education); Danielle Eisenberg (Ignite/Ed)
5	09.06.2026	How can professional development be ensured, and who has the responsibility and capacity to sustain it over time?	Prof. Russell Tytler (Deakin); Dr. Yael Granot-Bein (The Edmond de Rothschild Bridge)
6	21.07.2026	What is the right balance of physical, virtual and workplace environments, and how are industry mentors integrated?	Emlen Metz (UC Berkeley); Avi Salmon (Intel Israel)

The sessions ran for two hours on Tuesdays at 15:00 Israel time, a slot chosen so that participants in the United States, Europe and Australia could attend. Simultaneous Hebrew-English interpretation ran throughout, and every session was recorded.

From the third meeting the schedule was fixed and never moved again: fifteen minutes of opening remarks, two twenty-minute presentations each followed by five minutes of questions, forty minutes of open discussion, ten minutes of summary, and five minutes introducing the next dilemma. The two speakers were deliberately paired, one from Israel and one from abroad, and roughly a third of each session was unscripted discussion.

The first two meetings were not run that way. Meeting 1 was a convening session: participants introduced themselves in turn and one speaker gave a single thirty-minute lecture. The steering team's assessment afterwards explains the change: the open character of that session, combined with only partial use of the background materials circulated beforehand, had produced a broad and conceptual discussion that did not reach the depth available.

Roles held steady. Dr. Eisenberg convened, opened every session and introduced each following dilemma. Prof. Bentur set out the dilemmas and, from Meeting 4, moderated both question periods. Dr. Avigdor Zonnenshain summarised all six sessions. Dr. Revital Duek and Dr. Yael Granot-Bein moderated the open discussion from Meeting 3 onward. Tamar Dayan organised the meetings and built the document library on which this report depends. Ruchie Avital interpreted. The steering committee met seven times between 19 October 2025 and 4 August 2026.

2.3 The evidence base

This report is built from the Round Table's own documents: six meeting summaries; five programme decks, for Meetings 2 to 6; ten presenter decks and papers; two pre-circulated discussion documents, for Meetings 3 and 4; seven internal steering-committee summaries in Hebrew; three background papers circulated before the series; six published recordings; and the Phase I and Phase II reports. Appendix C lists them.

Positions attributed to named participants in sections 3 to 8 come from the meeting summaries. Statements about how the phase was planned and assessed come from the steering summaries, which are internal working documents and are cited rather than distributed.

2.4 Limits of the evidence

Four limits should be read before any attribution in this report is relied on.

Every meeting summary is a draft. All six were circulated for participant comment, and no final version of any of them was subsequently circulated. Where this report attributes a position, the attribution is to a circulated draft.

There are no transcripts. Every recorded position passed through simultaneous interpretation and then through a summariser's paraphrase into tidy reported prose. A hedged remark and a firm one look the same on the page.

Coverage is uneven. Meeting 1 is the thinnest session in the record: no deck for its single lecture, no programme deck, and a recorded discussion roughly a third the length of Meeting 2's. The participant lists are also not attendance records – Dr. Granot-Bein presented at Meeting 5 and speaks five times in its discussion, yet does not appear in that meeting's list.

The group largely agreed with itself. Phase III convened people who mostly shared a starting position, and the internal assessments of the sessions are close to uniformly positive. That is what made the discussions productive, and it is why section 9.4 records what the phase failed to settle at the same length as what it agreed.

Editorial note for the steering committee. The date of the first meeting is inconsistent in the sources. The invitation of 17 November 2025 and the schedule table reproduced in all five later programme decks give 02.12.2025; the meeting summary is headed 16 December 2025, which is also the date of a steering session that discusses the first meeting as already held. Table 1 uses 02.12.2025. Please confirm which is correct.

3. Meeting 1 – Skills and Competencies in the Era of Artificial Intelligence

3.1 The dilemma

There was none. The record states an objective in place of a dilemma: to shape the profile of lecturers and teachers for secondary and first-degree education in the generative AI era, with a focus on developing STEM competencies. Twenty-three participants and an interpreter were listed; eighteen introduced themselves and their work.

3.2 The presentations

Dr. Eli Eisenberg opened by placing Phase III against its predecessors: Phase I had identified the generic STEM competencies, Phase II had produced practical tools for educators and touched briefly on AI, and Phase III would extend the work to the profile of the educator, treating AI as a teaching instrument and as a cause of change in the competencies themselves. Prof. Arnon Bentur then presented the dilemmas that would structure the following meetings.

Avi Salmon of Intel Israel gave the session's single lecture, on the future educator in the age of AI. It distinguished three levels of engagement. At the first, everyone should be able to use the tools, and the decisive abilities are conducting a productive dialogue with a model and judging what comes back for truth, bias and appropriateness. At the second, professionals use AI to accelerate work they already understand, which fails without a strong professional foundation. The third, a working knowledge of how models are built and trained, was mentioned only briefly. The argument was that the future of teaching depends on educators reaching the second level, and that this cannot be acquired at second hand: a teacher cannot teach what the teacher has not experienced.

3.3 The discussion

Dr. Yael Granot-Bein named an asymmetry she encounters working with faculty. Educators are curious about their own use of AI, and the tone changes to suspicion as soon as students' use is raised; institutions then concentrate on controlling behaviour instead of reconsidering how they teach. Avi Salmon replied that resistance is usually a shortage of direct experience, and that the real issue is not cheating but understanding.

Dr. Einat Sprinzak put the most urgent need as basic AI literacy: people use these tools without grasping their limitations or biases, and the temptation is strongest for students because AI makes hard cognitive work feel easy. Telling students to use AI responsibly is not enough, and learning environments have to be designed deliberately for reflection and discernment.

Dr. Olena Bekh set the discussion inside a wider ecosystem in which the boundaries between formal and informal learning, and between digital and physical environments, are blurring. Two of her points recur through the whole phase: traditional assessment models no longer align with competency-based learning, and digital saturation can erode empathy, so emotional awareness has to be cultivated on purpose.

Danielle Eisenberg raised the load being placed on educators – expectations rising while professional development and institutional support do not – and asked how responsibility should be distributed between teachers, leadership and policy. Dr. Eisenberg's answer was that the change is not optional and that educators have to work with the tools themselves to stay relevant to learners. Prof. Bentur ended the round by rejecting the premise of a common profile: he does not believe all educators need to develop identical competencies, and what matters is that the system as a whole works.

3.4 Summary of the meeting

Dr. Avigdor Zonnenshain identified the emerging direction as a growing need to understand students better – their differing abilities, interests and ways of learning – and to move away from one-size-fits-all teaching.

The session closed by adopting the five dilemmas for the meetings that followed. Prof. Bentur's objection to a single profile became the dilemma of Meeting 2.

4. Meeting 2 – A Uniform Profile or a Diversity of Profiles

4.1 The dilemma

Should the profiles of lecturers and teachers become uniform or similar, or should there be a diversity of lecturer and teacher profiles within higher education faculties and secondary schools?

4.2 The presentations

Jan Morrison (TIES, United States) argued for a strategic reconstruction of the profession rather than an adjustment to it. Her proposal had two parts: recruit from varied backgrounds, industry included, as systemic strategy and not as a temporary fix; and build team-based models in which educators with different strengths complement one another. She named the orthodoxies that block this – teaching treated as a calling rather than as a profession that competes for top talent, and emerging technologies treated as optional rather than as part of teaching standards – and proposed replacing the single career path, which runs until burnout, with a road that has exits and re-entries. Her evidence-based examples were externships, in which teachers spend summers inside industry, and makerspaces used as instructional space from pre-kindergarten upward; her systemic example was Egypt’s STEM model schools.

Dr. Gabi Shafat (Afeka Academic College of Engineering) made the same argument from inside a single engineering faculty. He described AI as an apparently perfect lecturer – available at all hours, never impatient, faster at computation – and then named what it is not: confident when wrong, unaccountable when something fails in production, and incapable of human contact. He described his own advanced signal processing course rebuilt around teamwork, a flipped classroom, students teaching, and problem-based work, and reported the structural problem behind it: rising workload and role overload for engineering faculty, and a uniform lecturer role that no longer matches the discipline. His proposal was to move from a single-lecturer model to a complementary team, with unbundled academic roles, formal openness to recruiting industry professionals, and differentiated career paths. Engineering education, he concluded, should stop designing around the ideal individual lecturer, because diversity inside a well-designed system, rather than uniformity, will produce excellence.

4.3 The discussion

Asked by Dr. Zonnenshain what Afeka had done to demonstrate that any of this works, Dr. Shafat answered that the pace of change leaves no definitive answers yet. Take-home assignments have limited value once solutions rely on AI, so in-class assessment has to complement them, and that is not feasible in very large classes without systemic investment by policymakers.

Prof. Gil Noam questioned whether “teacher” and “lecturer” can be held in one profile at all, cautioning that the two work in significantly different contexts. Prof. Bentur argued for substantial commonality between secondary education and the first year or two of higher education, where student-lecturer ratios constrain instruction in the same way. Dr. Eisenberg raised academic freedom as a genuine difference, schools working under a prescribed curriculum while universities grant lecturers wide latitude; Prof. Bentur accepted the point for the third and fourth years. Dr. Marwa Maklada added that differences also run between middle school and upper secondary, and between STEM and humanities teachers. The question was left open.

Prof. Sigal Tifferet raised the evidence question that recurs throughout the phase: is embedding competencies inside disciplinary courses, rather than teaching separate skills courses, actually evidence-based? Her own review of the literature, she reported, did not reveal a clear consensus. Dr. Granot-Bein answered from UK and European research, which points to three conditions for competencies to be internalised – linkage to a discipline, repeated reinforcement rather than one-off activity, and structured reflection – and concluded that standalone skills courses attract lower participation and are perceived as less important. Dr. Maklada answered from Israeli programme results, including work validating competency-based tasks inside matriculation examinations.

Prof. Bentur named the structural cause: academic programmes have been collections of individual courses rather than pathways toward a defined graduate profile, and the change requires defining the graduate first and working backwards.

Avi Salmon connected the dilemma to hiring. Employers in high-tech are progressively less interested in routine technical ability that AI can perform and more interested in conceptual depth and genuine engagement with the profession. His point about the profession itself was blunter: the real question is not how to test students, but how to get the professors involved. Emlen Metz warned that momentum encouraging teachers to use AI has to be matched by equal emphasis on using it critically, and that students who edit AI output rather than produce their own work do not develop deep understanding.

4.4 Summary of the meeting

Prof. Russell Tytler closed the discussion round with a warning that increased student use of AI may lead students to relinquish control over their own thinking, and called for developing AI literacy and critical thinking together.

Dr. Zonnenshain's summary drew out three points: that education must go beyond transmitting knowledge and cultivate a love of the profession; that teaching should be team-driven, as engineering practice is, and teamwork therefore taught through collaborative work; and that industry involvement and project-based learning matter for motivation and for meaningful assessment. Dr. Eisenberg closed by identifying feedback, evaluation and assessment mechanisms as a major challenge across schools and higher education alike, and deferred the AI question to Meeting 4.

No profile was agreed and no vote was taken. The closest thing to a group position is the one stated by Morrison, Shafat and Bentur and left unchallenged: diversity inside a well-designed system, rather than a single ideal educator.

5. Meeting 3 – Possessing Competencies or Cultivating Them

5.1 The dilemma

Should lecturers and teachers themselves possess the required skills and competencies, or should their primary expertise lie in the ability to cultivate these skills among their students and pupils?

5.2 The presentations

Prof. Sigal Tifferet (Ruppin Academic Center) opened with the uncomfortable evidence on teacher training itself: across recent high-quality reviews, average effect sizes fall between 0.05 and 0.09. Small effects still matter at scale, but they set the standard for what counts as effective training. What works, on that evidence, is focused knowledge, explicit instruction with active learning and rehearsal, deliberate design for transfer through action plans and reflection, and organisational support. She proposed extending the familiar technology-pedagogy-content framework with an explicit skills axis, and taking each intersection seriously – so that critical thinking becomes argument analysis, taught through collaborative learning, supported by a specific digital tool. Her practical conclusion: for university teachers with no pedagogical training, concrete tools work better than abstract skills. She also raised modelling and its risk. Teachers should model the competencies they are developing, and where a teacher promotes a competency and behaves against it, the teaching loses credibility and may do harm.

Dr. Olena Bekh (European Training Foundation) presented the READY reference model for educators, built from a systematic review of the literature and of existing educator frameworks, then validated through workshops, with Israel and Ukraine as the first pilots. Its structure is six domains plus one, twenty-two professional practices and seventy-three descriptors; the summary names the domains as learner-driven, adaptability, sustainability and inclusion, collaboration and engagement, lifelong learning and reflection, and digital technologies. Her framing of what it is was as precise as the content: a reference model, selective and customisable, and explicitly not a qualification framework, not an appraisal instrument and not one-size-fits-all. The educator's role, on this account, moves from gate-keeper of knowledge to gate-opener.

Her answer to the dilemma was to refuse it. Deep understanding and practical experience are not alternatives but are role- and context-specific; teachers are not super-humans; building an educator's own competences and building the ability to teach them are equally important and mutually supportive. On the five competencies the Round Table works with – critical thinking, teamwork and interpersonal communication, self-directed learning, solving complex problems, and ethical judgment and responsible use of AI – she noted that none is specific to educators and that the list could and should be longer, since teachers are citizens too. On vulnerability she was direct: not knowing is a valuable driver of learning, and awareness of it grows as competence grows.

5.3 The discussion

Dr. Granot-Bein moved the question from pre-service training to the people already in the system: many experienced teachers are open to change but were never trained to teach competencies and cannot start from nothing. Prof. Tifferet held that the challenges are similar either way, and that introducing concrete tools is the realistic step for university teachers – provided the planning is done for them, so that educators implement rather than carrying both design and delivery.

Prof. Russell Tytler agreed that competencies should be taught concretely and then complicated the picture: many of them, creativity and problem solving among them, are context-dependent and do not decompose into fixed components, which makes translating an abstract skill into practice hard across disciplines. For experienced teachers, he argued, training should start from pedagogical content knowledge or it produces surface learning. Prof. Tifferet's reply left the question open: competencies probably have both general and context-dependent parts, and settling which is an empirical matter still unresolved.

Dr. Edith Manny-Ikan brought the classroom view. Teachers feel overwhelmed at having to teach extensive content and new competencies at once, and under time and assessment pressure they prioritise content. Change therefore has to include and support all teachers rather than the highly motivated ones. Prof. Bentur turned this into a systemic charge and put it to her directly: both schools and academia declare competency-centred graduate profiles, yet assessment practice does not reflect them, so how are competencies actually represented in the matriculation examinations? Her answer was a constraint rather than a solution – competency-based questions are perceived as harder and may disadvantage some students, so policymakers move cautiously and pilot gradually.

Dr. Duek reported a pilot at Tel Aviv University with Yale and the Edmond de Rothschild Bridge in which lecturers voiced the same concerns and by year's end mostly found the experience less difficult than expected. Dr. Granot-Bein went further: implemented properly, these approaches improve teaching rather than adding to it, but that requires a real shift away from frontal teaching, and it cannot rest on individual teachers. Dr. Maklada described a three-year national initiative involving hundreds of teachers, expanding through motivated lighthouse teachers, and attributed its progress to continuous support during implementation rather than to external solutions. The difficulties she named were variability between disciplines, thin capacity among teacher educators, and difficulty engaging school leadership.

5.4 Summary of the meeting

Dr. Duek closed the discussion with two points: the transformation has to be backed by policymakers and cannot be left to the field, and measurement, training and a clear implementation plan are all required. Dr. Zonnenshain put the group's task as demonstrating that competencies matter at least as much as content – an argument that looks widely accepted and, on the evidence of the day's discussion, is not. Dr. Eisenberg observed that teaching content has become easier over time while developing competencies remains a long process requiring patience and systemic adaptation.

The dilemma was not settled. Dr. Bekh's reframing – that possession and cultivation are not alternatives – is the nearest thing to a group answer, and no vote or agreed position is recorded.

6. Meeting 4 – The Educator, the Student and AI

6.1 The dilemma

Is the lecturer or teacher alone who should lead the educational process within their field of knowledge, or should an educational triangle be formed among the lecturer or teacher, the student, and artificial intelligence? Dr. Eisenberg opened by stressing the word “alone”.

6.2 The presentations

Dr. Iris Pinto (The Institute for Futures Research in Education) presented the method and findings of futures research: not prophecy, but the identification of trends and the choice of actions in the present. She set out four meta-trends she described as changing the nature of humanity – technological convergence, human-machine interfaces, human enhancement and the gap it opens, and life in a post-work world – and placed AI within them as an enabler that is only at its beginning.

Her deck answered the dilemma head-on, with a slide titled “There Is No Dilemma... There Is a New Educational Triangle”: AI neither replaces the teacher nor leaves the learner alone, the teacher leads pedagogically, the learner acts and explores, and AI expands, adapts and accelerates. She named the risks alongside it – over-reliance, weakened independent thinking, difficulty judging reliability, widening AI skill gaps between learners and lecturers, and blurred accountability – and the opportunities, chiefly personalised self-paced learning and more teacher time for guidance.

Danielle Eisenberg (Ignite/Ed) gave the opposing structural answer, that the triangle is not enough because the system is already layered. AI is not a tool inserted into a system but an agent that can act as content, tutor, evaluator and collaborator at the same time. The student-AI relationship is unavoidable and already happening with or without the school’s permission; the teacher-student relationship is irreplaceable; the teacher-AI relationship is underspecified, and it is the new professional role. She cited a 2025 meta-analysis finding that teacher-scaffolded AI use produces stronger outcomes while independent use produces dependency.

Her argument was that three failure modes are already observable in the absence of deliberate design: cognitive offloading, where higher confidence in AI means less critical thinking; erosion of the teacher-student relationship, with students reporting they feel less connected and prefer a tool that will not judge them; and hidden inequity. On the last she drew the sharpest line of the session. One-to-one tutoring is the strongest intervention education knows, so AI tutoring could be the most democratising force in its history; what is happening instead is that advantaged students use it to extend their thinking while disadvantaged students use it to replace theirs. The same tool, the opposite outcome, and a design failure rather than a technology failure. Her binding constraint was assessment: it measures outputs, so it cannot see shallow AI use, equity gaps, or whether learning happened at all, and every one of the three models loses that visibility without it.

6.3 The discussion

Dr. Noa Ragonis named teacher development as the recurring problem: educators are central to implementing any of this, and systems still fail to prepare and support them.

Dr. Revital Duek pressed three questions on the room – bring higher education and the lecturer into the triangle rather than discussing schools alone, and define what must remain human; how can a student’s thinking be made visible once AI is involved; and what design principles would make AI expand opportunity rather than deepen inequality. Prof. Gil Noam argued the opposite scoping, that the discussion should stay on K-12 and cover peer learning rather than teachers alone, and reported early findings that AI currently benefits teachers more than students, who often neither understand nor retain AI-generated work.

Prof. Tifferet proposed that if AI becomes genuinely effective at assessment, teachers could hand much of that work over – not as time management but as a redefinition of the role toward guidance, relationships and the human aspects of learning. Dr. Tzur Karelitz pushed further and asked why the teacher should mediate the machine’s evaluation at all. Danielle Eisenberg’s answer was that the limit is trust rather than capability: current systems may be biased because they are trained on data that does not represent diverse or neurodivergent students, so human oversight keeps the tools useful. Dr. Maklada held that the teacher must remain the architect of learning, responsible for deep thinking and for values-based education.

Answering Dr. Duek on visible thinking, Danielle Eisenberg described requiring students to demonstrate and defend their reasoning, including using AI to interview students about their own portfolios so that metacognition enters the cycle. Jan Morrison made the same point from the other end: students focus on what is measured, so presenting ideas and defending research should become central to assessment, as in the oral defence used in the Egyptian STEM schools. On equity her practical answer was procurement – the people choosing and buying AI tools should understand the technology, its purpose and the preparation teachers need before it reaches a classroom.

Avi Salmon put the state of the question plainly: AI is already an active participant, so what is left to decide is the teacher's role, and hands-on experience with AI has become a professional competency rather than an interest. Dr. Bekh warned that excessive dependence may reduce curiosity, creativity and empathy, and that losing critical thinking would mean losing part of what makes us human. Dr. Eisenberg raised the pace of change itself, asking whether more should be invested in researching the ethical and cognitive consequences before adopting further; Dr. Pinto replied that it does not depend on us, and that the useful work is understanding the possibilities.

6.4 Summary of the meeting

Dr. Zonnenshain summarised the session as a case for treating education as an integrated socio-technological system that includes teachers, AI, students, parents, communities and society, in which human roles remain central and teachers lead the learning process rather than only assessing it. Dr. Eisenberg closed by noting that AI requires careful and responsible consideration, and introduced the next dilemma.

The two presentations gave opposing structural answers – no dilemma and a new triangle, against a layered system in which the teacher-AI relationship is the undefined part – and the record does not reconcile them.

7. Meeting 5 – Who Sustains Professional Development

7.1 The dilemma

How can the professional development of lecturers and teachers be ensured? Who holds the responsibility and the capacity to sustain such development systematically and continuously over time – centres for advancing teaching and learning, a national centre for skills development, teacher educators?

7.2 The presentations

Prof. Russell Tytler (Deakin University) argued that developing an educator's STEM competencies is not the same problem at every stage of schooling, and that for secondary teachers the commitments and abilities largely arrive with the degree and the prior career. Teacher education's work is to convert them into classroom practice – pedagogical content knowledge. He presented research on STEM teacher identity showing that commitment in integrated STEM teaching comes less from disciplinary identity than from personal and pedagogical identity: curiosity, being a lifelong learner, wanting students to think critically. He presented a study of expert teachers guiding senior secondary research projects, in which the teachers intervene actively rather than waiting to be asked, using explanations, scaffolds, heuristics and

process constraints. And he presented his own faculty's undergraduate model: work-integrated learning across science degrees, with a core unit in which students source their own placements and complete more than eighty hours in research institutes or public agencies, and team projects delivered to real clients.

His final argument came from a primary classroom, where Grade 3 students designed a way to slow the melting of ice. Critical thinking there was not generic. It was carried in the discipline's own representations – annotated diagrams with arrows for heat exchange, causal statements grounded in observation, sequenced tables of measurements – which is why he holds that it has to be embedded in disciplinary practice rather than taught apart from it.

Asked about the Australian arrangement, he described professional development there as largely informal and decentralised, with responsibility resting on individual schools and no national framework, and said he does not regard it as an ideal model.

Dr. Yael Granot-Bein (The Edmond de Rothschild Bridge) presented the Israeli higher-education picture and the survey behind it: 90 per cent of faculty regard transferable competencies as highly important, 61 per cent have never received any formal training in teaching them, and 84 per cent of those not currently doing so are interested in learning how. The barriers named were training, time and institutional support.

Her central point was a chain that does not hold: understanding that competencies matter does not produce knowing how to teach them, and knowing how does not produce changed practice. She then set out the four candidate owners of the problem and what each cannot do alone. Individual faculty bring autonomy and motivation but reach mainly the innovators and do not scale – necessary and insufficient. Teaching and learning centres have the pedagogical expertise but limited authority and resources. Institutional leadership sets priorities and allocates resources, and without it faculty development rarely becomes institutional change. National infrastructure supplies shared language, shared tools and cross-institutional learning. Her line for the whole problem: research has national support structures, and teaching often does not.

She offered a working example in a national initiative spanning twenty-three institutions, in which teaching and learning centres act as local change agents inside a national community of practice. And she reported a finding about framing: faculty engage when transferable competencies are connected to academic excellence, research and disciplinary identity, and disengage when the same material arrives as labour-market rhetoric.

7.3 The discussion

Prof. Tifferet opened by naming a contradiction inside the system. Faculty are urged through workshops to integrate competencies while their institutions do not provide the resources, incentives or conditions to do it, and the result is frustration and burnout rather than change.

The national reform programme for Israeli higher education drew three assessments and no agreement. Prof. Bentur judged that it has significantly raised faculty awareness and is already driving change, since it requires institutions to define graduate profiles, redesign curricula, strengthen faculty development and build industry partnerships. Prof. Tifferet's understanding was that part of its funding came at the expense of earlier programmes that had already supported this work, in some cases at a smaller budget, so ambitious plans were never realised. Dr. Eisenberg held that it is too early to evaluate. Dr. Granot-Bein agreed the programme raised the national agenda while maintaining that responsibility for faculty development remains unclear at institutional level.

Dr. Duek asked the question the session did not answer: how do we know that professional development actually changed teaching and learning? Dr. Sprinzak's response came closest to a method – building development around a small set of specific teaching abilities, each with an answer to what it is, why it matters and how to do it better, plus tools for teachers to assess their own progress – on the grounds that teachers and lecturers are extremely busy and need to know what to do tomorrow.

Prof. Gil Noam asked for evidence comparing two approaches directly: teaching durable competencies as standalone skills, against embedding them in STEM subjects. He supports the connection and warned that the concept becomes vague without clearly defined competencies and explicit pedagogy. Dr. Eisenberg answered from programme experience rather than research, describing a national initiative that began four years ago with five pilot schools and now covers 353, evaluated annually, and drawing two lessons from it: embedding competencies in subject content is not enough, because they have to be made explicit, discussed and reflected on; and they have to be assessed, formatively and summatively.

Prof. Gil Noam also offered the phase's sharpest formulation of the human role, from medical education. As technology becomes more powerful, human relationships matter more, not less. Teachers will not disappear, but they will be valued less for transmitting knowledge and more for relationship and support – a paradox in which AI makes teachers less essential for some tasks and more important in their human role.

Prof. Tytler distinguished two models of development: teachers attending workshops that supply ideas, against a school bringing science, mathematics and technology teachers together to redesign curricula collaboratively. The second requires strong backing from school leadership and aims at a change in the school's culture. Dr. Karelitz argued that development should build confidence, adaptability and the capacity to keep learning rather than technical mastery of every new tool.

7.4 Summary of the meeting

Dr. Eisenberg closed the discussion with the point the session converged on: training educators is not enough on its own, and continuous accompaniment during implementation – guidance, mentoring and a supportive system as new approaches are put into practice – is the critical element if professional development is to change anything. Dr. Zonnenshain reinforced it from the national initiative's experience, describing a network of teachers of teachers who meet regularly and provide continuous guidance, and insisting that the support has to be practical: mentors need to work alongside teachers and observe their practice, because advice from a distance does not work.

8. Meeting 6 – Learning Environments, Industry and Experience

8.1 The dilemma

What is the appropriate balance between physical environments and virtual ones such as XR and VR? What forms of experiential learning in relevant workplaces are most suitable, and how can mentors from industry be effectively integrated into the learning process?

8.2 The presentations

Emlen Metz (University of California, Berkeley) set the two environments against each other on their merits. Virtual settings open possibilities, adapt easily, and are simpler to control and observe; they are also removed from reality and from other people, and offer shallower sensory experience. Physical settings offer real objects, face-to-face collaboration and full embodied experience, and are limited by available materials, space and staffing ratios.

The dangers were presented in two groups. Social erosion covers motivation, trust between student and teacher, academic integrity and social connectedness. Cognitive erosion covers literacy, cognitive offloading, critical thinking and what the presentation called a false sense of understanding – made worse, not better, by enthusiasm for the technology and by confidence in one's own ability to use it.

Against that, four productive uses: asking a model for counterarguments before a debate; translation into a student's language and reading level; active evaluation, the worked example being to ask an AI for its sources and then check those sources for reliability; and thinking first, brainstorming independently before consulting a model so that the model does not narrow the field. The presentation's design category, in its own words, was "AI-robust tasks" – laboratory work, building models, handwriting, and structured social interaction with defined roles. Its guiding principles were that difficulty and effort are necessary for learning, that practising is learning and students should not outsource a skill they have not yet acquired, and that some practice should remain deliberately AI-free.

Avi Salmon (Intel Israel) presented learning by experience from industry practice, through hands-on environments – makerspaces, hackathons, technical challenges and innovation hubs – in which participants learn by designing, building, testing and improving real solutions. The qualities he identified as decisive were curiosity, determination and humility. His observation from running these programmes was that participants who accepted uncertainty, explored independently and used AI as a support tool did considerably better than those who waited for instructions. Students learn more deeply as active participants, he argued – teaching younger peers, or working on authentic engineering projects – and talented students need room for independent work outside curriculum constraints.

8.3 The discussion

Dr. Duek opened with the question the session was built around: would learning environments look different if education had been designed around competencies rather than knowledge?

Dr. Gabi Shafat answered with the fullest worked example in the session. Afeka runs roughly twenty-four maker clubs across electronics, accessibility and music, with the college funding space, equipment and academic mentorship; the projects include a fuel-free electric vehicle designed to travel the length of the country and an autonomous marine vessel that competed internationally. The feature he identified as decisive was faculty involvement as mentors.

Dr. Noa Ragonis brought the evidence from the pandemic, when whether virtual laboratories could replace physical ones became a live question and the conclusion was that combining both works best. Virtual laboratories and digital tools improve accessibility and flexibility and do not replace physical experimentation, prototyping and engineering design. Dr. Sprinzak made the same point for simulations. Prof. Gil Noam objected to the framing of the choice itself: the question is not AI against hands-on learning but how AI can act as a collaborative learning partner – systems that ask questions rather than

supply answers, students collaborating while working with a model, and the teacher leading collective reflection. Emlen Metz and Avi Salmon both took up the practical form of this, comparing AI-generated arguments in class and prompting models to guide through questions, with discussion coming before the student turns to the tool.

Danielle Eisenberg named the obstacle that no one answered. The barrier to educational innovation is not a shortage of successful programmes but the inability to scale them, because assessment systems, university admissions and workforce expectations continue to reward traditional academic achievement rather than the competencies these models develop. The summary records no reply.

Prof. Bentur closed the discussion with the physical constraint. Interactive teaching is particularly hard in higher education because classes are extremely large, and reducing the student-to-faculty ratio is a condition for meaningful interaction rather than an improvement on it. He had raised the same constraint at Meetings 2 and 4. Again, no response is recorded.

8.4 Summary of the meeting

Dr. Zonnenshain reaffirmed learning by doing as a foundation for developing STEM competencies, with mentors and teachers acting as facilitators who guide rather than supply solutions, and identified makerspaces, fabrication laboratories and project-based programmes as effective settings.

Dr. Eisenberg closed the phase by naming the three themes that had recurred across all six meetings: that there is no single ideal profile for educators and that effective education may require diverse roles and complementary expertise; that educators need to develop the competencies they are expected to cultivate, particularly in AI and experiential learning; and that learning should increasingly integrate authentic workplace experience, industry mentors and hands-on activity. He set against them the balancing condition – that AI and virtual environments should complement rather than replace physical environments, collaboration and experience.

Jan Morrison proposed that the Round Table connect itself to other international round tables and professional networks, to exchange expertise and avoid duplicated effort.

9. Insights, Conclusions and Recommendations

9.1 Seven insights

Seven positions recur across the six meetings, and each was argued against before it held. Where the support is thinner than it looks, this section says so.

1. There is no single ideal profile. The foundation should be common and the expression diverse. Raised in five of the six meetings, and the only insight the group approached from every direction. Prof. Bentur stated it at Meeting 1 and never withdrew it; Jan Morrison and Dr. Shafat argued it independently at Meeting 2, from a national system and from one engineering faculty; Dr. Bekh restated it at Meeting 3 as flexible profiles, on the grounds that teachers are not super-humans. The cost is coordination, which nobody costed. A group of specialists that nobody coordinates is not a team.

2. Competencies are formed inside disciplinary learning, and the evidence for it is thinner than the consensus. The group converged on embedding rather than standalone courses, supported at Meeting 2 by European research on discipline linkage, repetition and structured reflection, and by Israeli programme results. But the same evidentiary gap opened twice and closed neither time. Prof. Tifferet reported at Meeting 2 that her literature review found no clear consensus; Prof. Gil Noam asked at Meeting 5 for evidence comparing standalone teaching against embedding, and warned that the concept becomes vague without defined competencies and explicit pedagogy. Both were answered from programme experience rather than research, and Phase I had already recorded that there is little evidence of skills transferring from one context to another.

3. Educators need lived experience of what they are asked to teach. Present in all six meetings, and the most repeated position of the phase. It should be recorded that it was carried mainly by one voice – Avi Salmon stated it in every session – with independent support from Dr. Eisenberg at Meeting 1, Dr. Sprinzak at Meeting 2 and Dr. Bekh at Meeting 3. What the meetings did not establish is how much experience is enough.

4. AI is already in the triangle. What remains to be decided is whether its role is designed or inherited. Meeting 4 produced two structural answers and reconciled neither: a new educational triangle in which the teacher leads, the learner explores and AI expands; and a layered system in which the student-AI relationship is already happening, the teacher-student relationship is irreplaceable, and the teacher-AI relationship is the new and undefined professional role. Both agree on the operative point. The default is not neutrality but a role assigned by accident.

5. Professional development has to include accompaniment during implementation, and no actor owns it. The strongest agreement of Meeting 5, supported independently by Dr. Eisenberg, Dr. Zonnenshain, Dr. Sprinzak and Prof. Tytler: workshops that precede change are not enough, and mentors have to work alongside teachers in real classrooms. Set beside it the ownership problem. Individual faculty do not scale, teaching and learning centres lack authority, institutional leadership is not sufficient on its own, and national infrastructure exists for research and rarely for teaching. Each holds part of the responsibility. None holds enough to act alone.

6. Physical, virtual and workplace environments should complement one another – and class size blocks most of what follows. Meeting 6 settled the first half quickly: virtual laboratories and simulations extend access and do not replace physical experimentation, prototyping and design. The second half is the more useful finding. Prof. Bentur named large class sizes as the constraint on interactive teaching at Meetings 2, 4 and 6, and on each occasion the record shows no response. It is the one structural obstacle the phase identified repeatedly and never addressed.

7. Assessment is the constraint that binds all the others. Named in every one of the six meetings and resolved in none. Dr. Bekh at Meeting 1: traditional assessment models no longer align with competency-based learning. Dr. Eisenberg's closing at Meeting 2: feedback and assessment mechanisms are a major challenge across the whole system. Prof. Bentur's charge at Meeting 3, that competency-centred graduate profiles are not reflected in assessment practice, was answered with an equity constraint rather than a solution. Meeting 4: assessment measures outputs and cannot see whether learning happened. Meeting 5: how would we know that development changed practice? Meeting 6: assessment systems, admissions and employers still reward what they always rewarded, which is why good programmes do not scale.

9.2 Guiding principles

Eight principles follow from the insights above. Each is traceable to positions taken in the meetings, and they are offered as a compass rather than as a specification.

1. Keep human judgment above the machine, and keep responsibility for learning human.
2. Treat AI as part of the learning ecology, designed in deliberately, rather than as a tool added beside it or a behaviour to be policed.
3. Keep disciplinary depth, and develop competencies inside the discipline rather than beside it.
4. Design for visible thinking rather than for polished outputs.
5. Build complementary teams of educators instead of pursuing one ideal profile.
6. Give educators lived experience of the practice they are asked to teach, before asking them to teach it.
7. Make professional development continuous, and accompany it through implementation.
8. Design for equity from the start, because the same tool produces opposite outcomes for advantaged and disadvantaged students.

Three of these pull against each other, and saying so is more useful than a flat list. Disciplinary depth and visible thinking compete for the same hours. Complementary teams and continuous development compete for the same people. Industry connection and equity conflict structurally, because partnerships form where partners already are.

9.3 Recommendations

The steering team required from the outset that recommendations be practical and separated by level of impact. Table 2 keeps that separation. The obstacle column is included because in every case the phase identified it and did not remove it.

Table 2. Recommendations by level of impact, and the obstacle named in the record.

Level	Recommendation	The obstacle
Personal	Require a short oral or written explanation of one submitted piece of work per unit, so that understanding rather than output is what is assessed	Marking time, with nothing removed to pay for it
Personal	Keep at least one deliberately AI-free task per unit, to protect the practice that the rest depends on	Nothing structural; it costs only the decision
Personal	Use AI as a source of counterarguments and as an object of evaluation, rather than as a source of answers	Requires the educator to have used the tools first
Institutional	Define the graduate profile first, map competencies across existing courses, and work backwards to the syllabus	Programmes are collections of courses, and mapping crosses departmental lines
Institutional	Fund accompaniment during implementation, not only the training that precedes it	Workshops are countable and cheap; implementation support is neither
Institutional	Put teaching development into promotion criteria, and collaboration time onto the timetable	Both are visible and expensive, and their returns are slow to attribute
Institutional	Reduce the student-to-faculty ratio where interactive teaching is expected	Named three times in the phase and costed never

Policy	Build assessment that can see inside the learning process, including at matriculation and admission	Competency-based questions are perceived as harder and raise real equity concerns
Policy	Create national support structures for teaching comparable to those that exist for research	Teaching has no equivalent of a national research infrastructure
Policy	Open recruitment to industry and other backgrounds as systemic policy, with differentiated career paths	Teaching standards and certification do not currently recognise them
Policy	Require equity guardrails and trained procurement wherever AI tools enter classrooms	Procurement decisions are made by people who are rarely trained for them

9.4 What Phase III did not settle

Four questions were raised repeatedly and closed by nobody. Reporting only convergence would misrepresent the phase.

Who pays, and who owns it. Every recommendation above carries a cost, and Meeting 5 established that no actor currently holds enough of the responsibility to act alone. Nothing in the phase was costed.

What comes out. Embedding competencies inside disciplinary courses means something already in those courses leaves. No meeting named what. Until someone takes a real syllabus and marks what would be cut, the recommendation is a request for time that nobody has.

Whether equity is served or damaged. This was raised as a warning and never as a measurement. Meeting 4 put a source behind the split between students who use AI to extend thinking and students who use it to replace thinking; Meeting 3 recorded that competency-based examination questions may disadvantage some students; Meeting 6 named the divide in basic access. No participant tested any of it against data from their own institution.

Whether any of it scales. The sharpest thing said in the phase, at the final meeting, was that assessment systems, university admissions and employer expectations still reward traditional academic achievement, which is why innovative programmes do not spread. It received no answer. The examination side was represented in the room throughout, and what was never present was the authority to change an examination, an admissions rule or a hiring screen.

9.5 Toward Phase IV

Phase III moved the question from the profile of the graduate to the profile of the educator. The four open questions above suggest that Phase IV would have to move it again, from the educator to the system that employs, assesses and rewards the educator.

Five themes were proposed for it. Three came from Dr. Eisenberg's closing remarks at the sixth meeting and were confirmed by the steering team on 4 August 2026: STEM leadership in the AI era, beyond technical expertise and across the continuum from school to work; the effect of continuing uncertainty and resilience on the competencies required of learners, teachers and professionals; and the relationship between policy set from above and innovation arising in schools, universities and professional communities. The steering team added two more: systems thinking and multidisciplinary, since STEM graduates can no longer rely on a single discipline; and the extension of the work from skills alone to values and attitudes, following the OECD's wider definition of competence.

Jan Morrison added a procedural proposal that belongs alongside them: connect this Round Table to other international round tables and professional networks, so that effort is not duplicated.

The steering team also recorded that continuation depends on funding, which was provided for Phase III by the Edmond de Rothschild Foundation and the Samuel Neaman Institute, and was not settled beyond it.

Appendix A – Round Table Participants

Roles and affiliations are given as the meeting summaries record them.

Name	Affiliation	Role in Phase III
Dr. Eli Eisenberg	Samuel Neaman Institute	Steering team, lead; opened all six meetings
Prof. Arnon Bentur	Technion	Steering team; set the dilemmas; moderated question periods
Dr. Avigdor Zonnenshain	Samuel Neaman Institute	Steering team; summarised all six meetings
Dr. Yael Granot-Bein	The Edmond de Rothschild Bridge for Higher Education and Employment	Steering team; presenter, Meeting 5; discussion moderator
Dr. Revital Duek	Policy Advisor on STEM	Steering team; discussion moderator, Meetings 3 to 6
Avi Salmon	Intel Israel, Innovation	Steering team; presenter, Meetings 1 and 6
Inna Zertser	Samuel Neaman Institute	Steering team; participant, all six meetings
Tamar Dayan	Samuel Neaman Institute	Steering team; organiser and source librarian
Golan Tamir	Samuel Neaman Institute	Participant, all six meetings
Jan Morrison	TIES – Teaching Institute for Excellence in STEM, USA	Presenter, Meeting 2
Dr. Gabi Shafat	Afeka Academic College of Engineering	Presenter, Meeting 2
Prof. Sigal Tifferet	Ruppin Academic Center	Presenter, Meeting 3
Dr. Olena Bekh	European Training Foundation	Presenter, Meeting 3
Dr. Iris Pinto	The Institute for Futures Research in Education	Presenter, Meeting 4
Danielle Eisenberg	Ignite/Ed	Presenter, Meeting 4
Prof. Russell Tytler	Deakin University	Presenter, Meeting 5
Emlen Metz	University of California, Berkeley	Presenter, Meeting 6
Dr. Marwa Maklada	Ministry of Education	Meetings 1 to 4
Dr. Noa Ragonis	Beit Berl Academic College	Meetings 1 to 4 and 6
Dr. Tzur Karelitz	National Institute for Testing and Evaluation	Meetings 1 to 5
Dr. Einat Sprinzak	Davidson Institute, Weizmann Institute of Science	Meetings 1, 2, 4, 5 and 6

APPENDIX A – ROUND TABLE PARTICIPANTS

Prof. Gil Noam	Harvard Medical School and PEAR Center	Meetings 1, 2, 4, 5 and 6
Dr. Edith Manny-Ikan	Henrietta Szold Institute	Meeting 3
Dr. Rinat Itzhaki	Henrietta Szold Institute	Meetings 4 to 6
Dr. Yair Noam	IDF Behavioral Sciences Department	Meetings 1, 2 and 5
Orly Rauch	Social Finance Israel	All six meetings
Dr. Merav Geva	Davidson Institute, Weizmann Institute of Science	Meeting 3
Suzan Hasan	JDC-TEVET	Meeting 3
Nadezda Solodjankina	European Training Foundation	Meeting 3
Oren Baratz	Jewish Federation of Cleveland	Meetings 1 and 3
Ruchie Avital	-	Interpreter, English-Hebrew-English, all six meetings

Editorial note for the steering committee. Several names are spelled inconsistently across the summaries, and this table picks one form for each. The Afeka presenter is spelled “Shaft” in the Meeting 2, 3 and 6 summaries and “Shafat” on his own title slide, which this report follows. The participant lists render Prof. Gil Noam surname-first as “Noam Gil”; he is a different person from Dr. Yair Noam. Danielle Eisenberg is unrelated to Dr. Eli Eisenberg, is given the title “Dr.” once in the record and not elsewhere, and describes her own work as spanning Ignite/Ed and ETS. Please correct any of these before publication.

Appendix B – The Six Meetings and Their Materials

- **Meeting 1**, 02.12.2025 (see the note in section 2.4). Introduction, participant presentations, and a lecture and discussion on skills and competencies in the era of artificial intelligence. A meeting summary and a published recording exist. There is no programme deck and no presentation file for the lecture.
- **Meeting 2**, 20.01.2026. Uniform or diverse educator profiles. Meeting summary, programme deck, both presenter decks, and a published recording.
- **Meeting 3**, 10.03.2026. Possessing competencies or cultivating them. Meeting summary, programme deck, both presenter decks, a published recording, and a discussion document circulated before the session.
- **Meeting 4**, 28.04.2026. The educator, the student and AI. Meeting summary, programme deck, both presenter decks, a published recording, and a discussion document circulated before the session.
- **Meeting 5**, 09.06.2026. Responsibility for professional development. Meeting summary, programme deck, both presenter decks, and a published recording.
- **Meeting 6**, 21.07.2026. Learning environments, industry and experience. Meeting summary, programme deck, both presenter decks, and a published recording.

- **Steering committee:** 19.10.2025, 16.12.2025, 03.02.2026, 24.03.2026, 12.05.2026, 23.06.2026 and 04.08.2026. Summaries exist for all seven.

Appendix C – Source Materials

Presentations. Jan Morrison, “Education System Perspective” (TIES), Meeting 2. Dr. Gabi Shafat, “The Challenge in Engineering Higher Education”, Meeting 2. Prof. Sigal Tifferet, “How to train teachers to impart skills?”, Meeting 3. Dr. Olena Bekh, “Educators at the Times of New Learning”, Meeting 3. Dr. Iris Pinto, “Future Trends Shaping Reality”, Meeting 4. Danielle Eisenberg, “Default or Design – Why the Teacher’s Role in the AI Era Is a System Architecture Problem”, Meeting 4. Prof. Russell Tytler, “STEM skills for teachers”, Meeting 5. Dr. Yael Granot-Bein, “Who Sustains Faculty Development for Transferable Skills? Lessons from Israeli Higher Education”, Meeting 5. Emlen Metz, “Balancing In-Person vs. Computer-Based and AI Experiences in Education”, Meeting 6. Avi Salmon, “Training by experience”, Meeting 6.

Round Table documents. Six meeting summaries, one per session. Five programme decks, for Meetings 2 to 6. Two discussion documents, both titled “The Teacher in the AI Era – A Discussion Document”, prepared for Meetings 3 and 4 by Dr. Eli Eisenberg, Prof. Arnon Bentur and Avi Salmon. Seven steering-committee summaries, which are internal working documents in Hebrew and are cited in this report rather than distributed.

Prior phases. *International Round Table Phase I for Advancing Skills in STEM Education*, Samuel Neaman Institute, January 2025. *International Round Table for Advancing Skills in STEM Education, Phase II*, Samuel Neaman Institute, March 2025.

Background materials circulated to participants. *OECD Teaching Compass: Reimagining Teachers as Agents of Curriculum Change*, 2025. *Results from TALIS 2024: The state of teaching through the eyes of teachers*, OECD, October 2025. *Envisioning Educator Roles for Transformation*.

Frameworks named in the presentations and discussion documents. The OECD Learning Compass 2030; the UNESCO AI competency frameworks for teachers and for students; the EU key competences for citizens (2018); the ETF READY reference model, the Scaffold card deck developed with the European Commission’s Joint Research Centre, SELFIE and SELFIE for Teachers, and the FormA formative assessment toolkit.